

SPECTRE-1 Scratch Storage System

A 150–250 PB classified scratch file system for Oak Ridge National Laboratory, transitioning to the Department of War under RMF. This brief covers the solution recommendation, the constraints that decide the award, every edge case we found in the package, and — in Appendix A — everything we ruled out and why.

CUSTOMER UT-Battelle, LLC / ORNL	END USER "The Sponsor" → DoW	SITE Bldg 5300, DC-1N Oak Ridge, TN	DELIVERY Q1 CY2027
ACCEPTANCE Q2 CY2027	FULL PRODUCTION Dec 2027	SOURCE Attachment C v0.4 May 2026	

01 Bottom line

Bid a **hybrid flash + CMR-HDD parallel file system**. Not all-flash, not all-HDD.

The competition will not be decided on bandwidth. It will be decided on four physical and regulatory constraints that most OEMs' standard 2027 configurations violate on day one, and on a set of non-technical gates that eliminate vendors regardless of how good the array is.

RANK	SOLUTION	FILE SYSTEM	WHY IT FITS
1	DDN EXAScaler AI400X3/X3M flash SSU + 4U102 HDD SSUs	Lustre	Native POSIX, bare-metal control plane, deepest cleared DOE/DoD field-service bench. AI400X3M added hybrid flash+HDD and 30 PB/rack at ISC 2026.
2	IBM Storage Scale System SSS 6000 flash + HDD expansion	GPFS	Stores files $\leq \sim 3.5$ KiB <i>inside the inode</i> — a direct architectural hit on the 40.2% of files that are ≤ 4 KiB. Nobody else has a cleaner answer to this distribution.
3	HPE Cray Storage Systems E2000	Lustre	Native Slingshot. ORNL already runs Orion — a 679 PB ClusterStor Lustre file system — with the same operations staff who will run this one.
4	VDURA V5000 + WD Data102	PanFS	VeLO director nodes hold metadata <i>and</i> small files on NVMe; VPOD/HDD carries bulk. 2026 roadmap targets ~ 200 PB usable at 2.5 TB/s.

RANK	SOLUTION	FILE SYSTEM	WHY IT FITS
5	WEKA	WekaFS	Best raw small-file and metadata IOPS in the field, with a true POSIX client. Capacity economics are the problem, not the engine.

Everything we considered and rejected — six platform categories and four architectural options — is documented in **Appendix A** with the reasoning, so the decisions survive a review six weeks from now.

02 Deal shape

Attach. F

The procurement is deliberately **split into two fixed-price subcontracts**, and Attachment C repeatedly marks sections *"to be quoted by the OEM"* (§3.1.8, §5, §8.4).

	HARDWARE & INSTALL	OEM MAINTENANCE & SUPPORT
Period of performance	10/31/2026 - 10/30/2032	12/15/2026 - 12/14/2032
Type	Commercial Items, Fixed Price	Commercial Items, Fixed Price
NAICS	541519 — IT Value Added Resellers Size standard: 150 employees	541519 — Other Computer Related Services Size standard: \$34M

WHY THIS MATTERS TO US

The hardware subcontract carries the **ITVAR NAICS with a 150-employee size standard**. That is the single most important commercial fact in the package for a value-added reseller. The structure invites a VAR-primed hardware award paired with an OEM-primed maintenance award, with an explicit handoff written into §8.5: *"After acceptance tests conclude, a handoff of system responsibility will occur from the hardware awardee to the maintenance/support awardee."*

Confirm the small-business set-aside status early — it determines whether this is a direct bid or a teaming play.

Schedule

FACILITY READY	DELIVERY	ACCEPTANCE
Before Q4 CY2026	Q1 CY2027	Q2 CY2027

LIMITED PRODUCTION	FULL PRODUCTION	EARLIEST DECOMMISSIONING
Q3 CY2027	December 2027	Q2 CY2031

03 The four constraints that decide this

3.1 No HAMR, no MAMR, no SMR — this caps HDDs at 26 TB

§3.2

"When applicable, use PMR/CMR technology for all HDDs; HAMR, MAMR, and Shingled recording devices are not permitted."

This is the highest-impact line in the document and it is easy to miss.

- Seagate's 30 TB+ Mozaic drives are **HAMR** — excluded.
- WD's 32 TB / 40 TB-class drives are **UltraSMR** — excluded.
- Toshiba's high-capacity parts are **MAMR** — excluded.
- The largest compliant drive is the **WD Ultrastar DC HC590 at 26 TB** (ePMR/CMR, 11-platter, SAS or SATA, 2.5M-hr MTBF), with 24 TB as the safe volume part.

Every OEM's default 2027 capacity-optimized quote will assume 30–40 TB drives. Under this rule they lose roughly **35-50% of the capacity per rack-unit and per watt** they were counting on. Any competitor quoting HAMR or SMR is non-compliant; any who requotes on CMR watches their density collapse.

OPEN — RESOLVE BEFORE SIZING ANYTHING

Confirm a **FIPS-certified SED SKU exists at 26 TB**. The HC590 is documented with Instant Secure Erase, but a FIPS 140-2-certified or 140-3-compliant SED variant at top capacity is *not* confirmed, and the SED/FIPS SKU historically trails the leading capacity by a generation. If it does not exist, the compliant ceiling drops to 24 TB or lower and **every capacity number in this brief moves with it**.

3.2 Power, not floor space, is the binding constraint

§7.1

The electrical spec quietly sets the whole system envelope.

- 3-phase **60 A / 208 VAC** receptacles → 21.6 kVA, derated 80% continuous = **17.3 kW usable per feed**.
- Dual-fed **diverse** — one utility, one UPS. You must survive on *one* side, so design to 17.3 kW per rack.
- Total cap **275 kW** across production, cluster management, and the 5300-DC TDS.

- 275 kW ÷ 16 production racks ≈ **17.2 kW/rack**. The two numbers are the same constraint stated twice — ORNL sized the budget at exactly one 60 A feed per rack.

Cooling is *not* the limit: an RDHX at 11 gpm with a 15 °F delta removes roughly 24 kW, comfortably above the 17.3 kW electrical ceiling. **Design to watts, not to BTUs or to rack units.**

And because §3.2 requires the base proposal to occupy only **50% of production floor space = 8 racks**, the base bid has a hard budget of roughly **138 kW**.

3.3 What 138 kW actually buys

A WD Ultrastar Data102 4U102 JBOD draws **~1,300 W typical** (1,600–1,800 W max). Working the numbers against the 8-rack base bid:

MEDIA STRATEGY	MAX USABLE IN 8 RACKS / 138 KW	400G ENDPOINTS AFFORDABLE	VERDICT
All-HDD (26 TB CMR)	~184 PB	~20	Capacity ✓ · bandwidth-starved ✗ · small-file ✗ ✗
All-flash (61.44 TB QLC)	~250 PB	~48–64	Best fit for the envelope — but cost and NAND lead time ✗ ✗
Hybrid RECOMMEND	~150–180 PB	~30–40	Balanced; lands at the low end of the required band

TWO CONCLUSIONS WORTH PUTTING IN FRONT OF ORNL

1. **All-flash fits the facility envelope *better* than all-HDD.** Spinning disk is what blows the power budget — 87 JBODs consume 114 kW and leave almost nothing for the servers carrying the 400G endpoints. This is counterintuitive, and it is the strongest argument for a flash-heavy hybrid.
2. **In 8 racks you land near 150 PB — the bottom of the 150-250 PB band.** Reaching 250 PB requires either the full 16 racks or an all-flash build. See Q1 in §06.

3.4 The file distribution is a metadata problem, not a capacity problem

§4.1

Converting the file-count distribution to bytes using log-midpoint estimates:

BUCKET	% OF FILES	≈ % OF BYTES
≤ 4 KiB		40.2% ~0.003%
≤ 32 KiB		24.0% ~0.016%
≤ 64 KiB		8.1% ~0.014%
≤ 256 KiB		9.7% ~0.06%

BUCKET	% OF FILES	≈ % OF BYTES
≤ 1 MiB		3.8%
≤ 4 MiB		9.8%
≤ 16 MiB		1.1%
≤ 1 GiB		2.9%
> 1 GiB		0.5%

72.3% of files consume ~0.03% of the bytes. Mean file size lands around 25–30 MiB, which at 200 PB implies roughly **4–10 billion inodes**. The estimate is sensitive to the ≤1 GiB bucket midpoint — plan for 10 billion.

CORE TECHNICAL WIN THEME

A flash tier of only **~0.5–1 PB usable** holds every inode plus every file ≤ 64 KiB — absorbing 72% of all file operations — while HDD carries 99.97% of the bytes. Small files are cheap to solve here. Spend the flash budget on metadata and the small-file population, and the rest on capacity.

This is also where the file systems genuinely differ:

- **IBM Storage Scale (GPFS)** stores file data *inside the 4 KiB inode* when it fits (~3.5 KiB payload). The 40.2% of files that are ≤4 KiB require **zero data-block I/O** — the read or write completes in the metadata operation itself.
- **Lustre** needs **Data-on-MDT** plus PFL and DNE2 striping, with flash MDTs, to reach equivalent behavior. It works, and DDN and HPE both know how to tune it, but it is configuration rather than architecture.
- **VDURA/PanFS** routes small files to the VeLO director nodes on NVMe by design.
- **WEKA** distributes metadata across the cluster and is the strongest performer of the group on pure small-file IOPS.

04 Reference configuration — base bid, 8 racks

LAYER	CONTENT	CAPACITY	POWER
Metadata + small-file tier	~8 × 2U24 NVMe TLC, FIPS SED (30.72 TB class)	~0.6 PB usable	~12 kW
Performance flash pool	61.44 TB QLC E1.L, FIPS 140-3 Level 2 capable	~25 PB usable	~15 kW
Capacity tier	~60 × 4U102, 26 TB ePMR CMR SED	159 PB raw ~127 PB usable @ 8+2	~78 kW
Endpoint / server layer	~30 servers × 400G (Slingshot / UE / IB)	—	~36 kW
Management + TOR switching	2 TOR per rack (customer-provided network)	—	~8 kW
Total	~306U of 400U available	~152 PB usable	~137 kW

The remaining 8 racks are priced as SSU growth options per §3.2, taking the system to roughly 300 PB usable at full build.

Priced separately, per the RFP

- **2 × TDS** — each a minimal but complete instance: metadata server, one flash SSU, one HDD SSU, and management. One occupies the 17th rack at 5300-DC.
- **On-site parts cache** sized from MTBF to sustain two weeks with no refresh, including hot spare nodes of every type.
- **1 × on-site System Software Analyst, DOE Q-cleared**, four years plus 5th/6th-year options.
- RHEL licensing and support, file system licensing, HA licensing.

DESIGN LEVER MOST BIDDERS MISS

Prefer small SSU granularity, and minimize the number of distinct node types.

Two TDS systems each requiring "a minimal set of each component" is a fixed cost that scales with how coarse the building block is — and \$5.2 separately requires *hot spare nodes of each type* in the parts cache. Every additional node type is paid for twice. A vendor whose minimum viable unit is a 2U flash + 4U disk pair prices both line items far better than one whose minimum unit is a populated rack.

05 Edge cases

Everything in the package that will cost money, time, or a protest if it is discovered late. Grouped by domain; severity in the left column.

CHIP MEANINGS BLOCKER Stops the bid or the design until resolved	VERIFY Get written confirmation from the OEM or vendor	ASK Submit as a clarification question to ORNL
WATCH Price it, plan it, don't be surprised by it		

REF	EDGE CASE	CONSEQUENCE
CAPACITY & DEFINITIONAL		
ASK	\$3.2 / \$3.2.2	Does 150-250 PB apply to the 8-rack base bid or the full 16-rack build?
ASK	\$3.2.2	Is "PB" decimal (10 ¹⁵) or binary (PiB)?
		Moves the bid by a factor of two. Under the strict reading, ~150 PB is the honest answer and 250 PB is unreachable without all-flash.
		12.6% swing on every capacity number. If PiB, the 8-rack case fails outright.

	REF	EDGE CASE	CONSEQUENCE
ASK	§3.2.2	150–250 PB is a <i>range</i> — it has a ceiling as well as a floor.	Building out the remaining 8 racks with SSU options per §3.2 pushes past 250 PB. The two requirements contradict each other at full build.
ASK	§3.2.2, §4.2	How is "usable" defined — post-erasure-code, post-filesystem-overhead, or post-aging?	Acceptance runs at 70% utilization of a namespace "formatted the same as production." Whether the 70% is measured against raw or usable changes the delivered figure.
MEDIA			
BLOCKER	§3.2	HAMR / MAMR / SMR all barred — 26 TB CMR ceiling.	Invalidates the density assumptions in essentially every vendor's 2027 capacity configuration.
BLOCKER	§3.2	FIPS SED required on <i>all</i> media — and the SED SKU may not exist at 26 TB.	Drops the compliant ceiling to 24 TB or lower and moves every capacity number in this brief.
VERIFY	§3.1.6	TAA and hard drives. Most enterprise HDDs are manufactured in Thailand and Malaysia — neither is a TAA-designated country .	The most commonly litigated compliance issue in federal storage. Usually addressed through substantial-transformation claims. Get a written country-of-origin statement per part number; do not assume.
VERIFY	§3.1.6	TAA and NAND. Several high-capacity QLC lines are assembled outside TAA-designated countries.	Can eliminate the specific drive that made the all-flash or hybrid math work. Verify before the config is locked.
WATCH	§3.1.5, §3.2	FIPS 140-2 certificates moved to historical status in 2026, yet §3.2 still accepts "FIPS 140-2 certified" SEDs.	A drive with a historical-status certificate may or may not satisfy the assessor. Worth an explicit question.
VERIFY	§5.1, §4.2	QLC endurance against a <i>scratch</i> workload for a 4–6 year PoP.	Scratch is write-heavy and churning; QLC DWPD is low. Model projected daily turnover against the warranty, or put TLC on the write-intensive tier.
WATCH	§3.2	ECC DIMMs of a uniform FRU type across the entire system.	One DIMM part number everywhere. Metadata servers normally want more RAM than storage servers — this forces over-provisioning on some nodes.
POWER, FACILITY & MECHANICAL			
BLOCKER	§7.1	17.3 kW per rack on a single feed; ~138 kW for the 8-rack base.	A design that assumes both feeds are simultaneously usable (34.6 kW) loses the utility/UPS diversity the spec requires.
ASK	§7.1	Is the UPS side sized for full load, or only for orderly shutdown?	Determines whether the system must ride through or shut down cleanly on a utility event — which changes the HA and write-cache design.
ASK	§7.1, §7.2	Do RDHX fans count against the 275 kW?	Fan-assisted RDHX adds 1–2 kW per rack. At 17 racks that is up to 34 kW off an already tight budget.
VERIFY	§7.2	Passive RDHX at 17 kW/rack with 60–68 °F supply water and a 75 °F inlet is a tight approach temperature.	May force an active (fan-assisted) RDHX — which loops straight back into the power budget above.
ASK	§7	No floor loading limit is stated anywhere in the package.	A 600 mm × 1300 mm cabinet with eleven fully populated 4U102 JBODs exceeds 1,400 kg. On-grade slab, no raised floor — the rating must be confirmed before cabinet selection.

	REF	EDGE CASE	CONSEQUENCE
WATCH	\$7	52" (1321 mm) freight aisle vs. permitted 1300 mm extra-deep cabinets.	Roughly 20 mm of clearance. Transit is feasible in a straight line; turning radius at corners is the real problem. Walk the route before committing to deep cabinets.
WATCH	\$7	No seismic or anchoring specification given.	Confirm before finalizing cabinet and mounting hardware.
WATCH	\$7.2	<5% parasitic heat to the room during short-term maintenance.	At 275 kW that is 13.75 kW into the data hall. Constrains how many RDHX circuits can be open at once, and therefore maintenance procedure design.
WATCH	\$7.2	RDHX leak detection with automatic shut-down capability.	A leak event shuts the control valve and strands that rack thermally. The file system needs graceful degradation for a rack going hot, not just for a rack losing power.
BLOCKER	\$7.2	The RDHX scope is mechanical contracting, not storage: 1in sanitary connections, top-fed, fail-closed valves, BACnet/IP, check valve on return, control valve on supply, plus delivery, uncrating, staging, mounting and connection.	No storage OEM owns this competency. Identify and price a mechanical partner before bid.
WATCH	\$7	At most two top-of-rack management switches per rack, on a separately procured customer network.	Count BMC, JBOD expander, PDU, and RDHX BACnet ports per rack against that budget. Coordinate VLANs, addressing and DHCP with ORNL early.
WATCH	\$3.1.4	Inter-cabinet cabling must be optical or shielded twisted pair.	Rules out passive copper DAC between racks. Budget AOC/optics across the entire inter-rack fabric — a real line item at this port count.
ASK	\$5.2, \$7	Does the burn-in / diagnostic cabinet count against the 17-rack limit?	\$7 counts cluster management and the TDS explicitly; the diagnostic cabinet is mentioned only in \$5.2 and left ambiguous.
ARCHITECTURE & FILE SYSTEM			
ASK	\$3.2.2	Does "POSIX-compliant namespace" require a native POSIX client, or is POSIX semantics over NFSv4.1/RDMA acceptable?	Determines whether VAST-class solutions are in scope at all. See Appendix A.
ASK	\$3.2	What is SPECTRE-1's compute fabric — Slingshot, Ultra Ethernet, or InfiniBand?	Slingshot is HPE-proprietary and materially advantages HPE. UE or 400 GbE levels the field. Also gates third-party NIC availability.
WATCH	\$3.2	Ultra Ethernet silicon is early-generation in 2026–27.	If quoting UE, state firmware maturity and interoperability evidence, and offer 400 GbE as the low-risk alternate.
WATCH	\$3.2.2	"Up to 120 endpoints" is a ceiling, not a floor — and power caps it far lower.	Each 400G server costs ~1.1–1.2 kW against a ~138 kW budget. If ORNL expects 120 endpoints, the power budget cannot support it with HDD. Surface the trade rather than silently under-delivering.
WATCH	\$3.2.3	Container-orchestrator preference is stated as a preference, not a mandate — and the sentence is truncated mid-clause in v0.4.	Favors Lustre and GPFS. VAST and WEKA need a crisp written rebuttal rather than an assumption that it doesn't apply.

	REF	EDGE CASE	CONSEQUENCE
WATCH	§3.3	Home directories live on a separate, existing file system provided by ORNL.	Scratch must interoperate cleanly with it. Confirm mount topology and identity mapping.
ASK	§3.3	ORNL provides RSA, LDAP, DNS and license management — but no KMIP key manager is mentioned.	SEDs at rest need key management. Establish who owns it before assuming either side.
SECURITY & COMPLIANCE			
BLOCKER	§3.1.11	All personnel — staff, vendors <i>and subcontractors</i> — engaged in design, development, integration or operation must be US citizens.	Broader than most vendors' standard "US persons for on-site work" posture. Rules out any OEM with offshore engineering anywhere in the delivery path.
BLOCKER	§5.5	On-site System Software Analyst requires a DOE Q-clearance .	Sponsorship and lead time. Very few storage OEMs have cleared analysts to spare.
VERIFY	§3.1.10, §5.4	DISA STIG compatibility <i>and</i> RHEL licensing — but many storage vendors forbid customer OS modification.	There is no DISA STIG for Lustre or GPFS, so the RHEL STIG gets applied to the servers. Get a written support statement that applying the RHEL STIG does not void support — this eliminates appliance platforms that lock the OS.
VERIFY	§5.4	RHEL specifically is required — not a derivative or a vendor Linux.	Some HPC storage appliances ship a vendor-specific or SLES-based OS. Confirm the platform is genuinely RHEL.
WATCH	§3.1.1	NIST SP 800-207 Zero Trust applied to a parallel file system data path.	Parallel file systems generally trust the client. Lustre without Kerberos/GSS trusts it entirely. There is real architectural tension here — and GSSAPI carries a performance and operational cost. Ask how ZTA is expected to apply.
VERIFY	§3.1.1	NIST SP 800-193 firmware resiliency across <i>all</i> components.	Root-of-trust and firmware recovery on drives, HBAs, NICs, expanders, PDUs and RDHX controllers. Many JBOD expanders and intelligent PDUs do not meet it.
WATCH	§3.1.1	NIST SP 800-223 (HPC Security) is recent and few storage vendors have mapped controls to it.	A differentiator for whoever does the mapping work — and a risk for whoever assumes it is covered by 800-53.
BLOCKER	§3.1.8	Critical/High CVSS v4 fixes available <i>and applied</i> within 30 days — including third-party dependencies.	A hard contractual SLA in an air-gapped classified enclave. Requires a real SBOM practice plus a defined media-transfer process. Note the RFP specifies CVSS v4 while most vendor advisories are still v3.1.
WATCH	§3.1.13, §3.1.14, §3.1.9	The documentation package: SCIS for all 800-53 High controls, SSP, CM Plan, IR Plan, SCRM policy and SOP, supplier risk register and scorecards, authenticity logs, firmware integrity scans, SDLC checklists, secure baselines, sample SOWs, SCAP reports, audit logs, config checklists.	Hundreds of professional-services hours. Price it explicitly — do not absorb it into the hardware margin.
WATCH	§3.1.12	The secure SDLC requirement attaches to <i>any</i> system software development or integration.	Even modest custom integration work triggers the full artifact set.
ACCEPTANCE			

	REF	EDGE CASE	CONSEQUENCE
BLOCKER	\$4.2	Acceptance runs against a namespace artificially aged to ~70% utilization , with IOR and mdtest compared directly against published specifications.	The single largest technical risk in the package. Every vendor datasheet number comes from an empty file system; aged metadata performance can be materially lower.
ASK	\$8.5	The aging methodology is deferred to contract negotiation.	File size mix, create/delete churn and directory fan-out swing mdtest by large factors. Defining this is the highest-leverage negotiation item in the entire package.
WATCH	\$8.5	Benchmarks must be executable from a <i>non-privileged</i> account.	Tuning that requires privilege — dropping caches, setting stripe parameters — cannot be part of the demonstrated run.
WATCH	\$8.5	Injected failures during acceptance, plus a stability test holding the system fully subscribed.	Rebuild performance on 26 TB drives under sustained load is a genuine risk. Declustered RAID matters far more here than in a datasheet comparison.
WATCH	\$4.2	Physical validation of "foam kits, screw installation, and properly applied torques."	Unusual, and it means documented torque specifications and installation QA are acceptance artifacts.
COMMERCIAL & LOGISTICS			
BLOCKER	\$5.3	No hardware leaves the site — ORNL destroys it.	Every FRU, every drive, for six years, with no failure-analysis returns. Requires a full hardware-retention service priced across the entire BOM. Some OEMs will not warranty without returns.
WATCH	\$5.2, \$5.3	No RMA feedback loop means no way to refine MTBF estimates in service.	The parts cache must be sized conservatively from the start, because it can never be tuned from observed failure data.
WATCH	\$5.2	Spare parts remain Offeror property until introduced into the system — but no hardware leaves the site.	Unused spares at contract end are stranded and destroyed. That loss belongs in the bid, not in the post-mortem.
WATCH	\$5.2	The cache must sustain <i>all</i> repair actions for two weeks with no refresh — and include hot spare nodes of every proposed type.	Infant mortality dominates the first year and runs well above steady-state AFR. Size for the first year, not the average.
ASK	\$3.2.1, \$7	Where does the second TDS live? \$7 accounts for only the 5300-DC TDS in the 17-rack footprint.	Another ORNL location, an unclassified enclave, or a vendor site — and whether it must be cleared. Changes site-work pricing.
ASK	\$3.2.1	TDS support co-terminates with the production <i>compute</i> partition, not with this storage contract.	The support end date is tied to something outside this procurement. Confirm the date before pricing the line.
ASK	\$5.1, Attach. F	Maintenance PoP starts 12/15/2026, but \$5.1 says maintenance begins upon Acceptance (Q2 CY2027).	Roughly two quarters of ambiguity in a fixed-price support quote.
WATCH	\$1, \$5.1	Earliest decommissioning (Q2 CY2031) precedes the end of the maintenance PoP (12/2032).	Confirm what the 5th and 6th year options actually cover if the system can be decommissioned first.
WATCH	Attach. F	DOE O 142.3B — all deliveries to ORNL must be made by a US citizen driver . A foreign national will be turned away at the gate.	Standard freight carriers do not guarantee this. Contract it explicitly, or a delivery slips.

	REF	EDGE CASE	CONSEQUENCE
WATCH	Attach. F	Activity Hazard Analysis required and approved <i>before</i> work performance (OSHA 1910/1926, 10 CFR 850, NFPA 70E).	Gates the Notice to Proceed. Start it early; it is not a formality.
WATCH	§6, §7.2	Offeror uncrates, inspects, relocates, stages, mounts and removes all packaging — but <i>ORNL</i> physically completes the connections to Offeror equipment.	Split responsibility at the final connection. Clarify liability for a fault introduced at that boundary.
WATCH	§1	The RFP explicitly invites alternative delivery schedules to mitigate long-lead risk.	A phased delivery — metadata and flash tier first, capacity tier second — protects the Q2 CY2027 acceptance date. Offering it is free; needing it later is not.

06 Clarification questions to submit

Ordered by how much the answer changes the bid.

- Q1 Does the 150-250 PB usable requirement apply to the 8-rack base proposal, or to the full 16-rack build-out?** Under the strict reading the answer is ~150 PB — the very bottom of the band — and 250 PB is unreachable without all-flash. This single answer moves the bid by a factor of two.
- Q2 Is "PB" decimal or binary?** If PiB, every capacity number rises 12.6% and the 8-rack case fails.
- Q3 Is a FIPS-certified SED required at 26 TB, and will ORNL accept a lower drive capacity or filesystem-level encryption if no FIPS SED SKU exists at that capacity?**
- Q4 Does "POSIX-compliant namespace" require a native POSIX client, or is POSIX semantics over NFSv4.1/RDMA acceptable?**
- Q5 What is SPECTRE-1's compute fabric — Slingshot, Ultra Ethernet, or InfiniBand?**
- Q6 Where does the second TDS live, and does it require a cleared facility?**
- Q7 Who provides key management for the self-encrypting drives?** §3.3 lists RSA, LDAP, DNS and license management as Company-provided but omits a KMIP key manager.
- Q8 Will ORNL define the namespace-aging methodology for acceptance before award?**
- Q9 What is the floor loading limit for DC-1N?** No value appears anywhere in the package.
- Q10 Do RDHX fans and controls count against the 275 kW cap?**
- Q11 Is the UPS-protected feed sized for full load or orderly shutdown?**
- Q12 Can the maintenance PoP start date (12/15/2026) be reconciled with §5.1's "begins upon successful completion of Acceptance"?**
- Q13 How is NIST SP 800-207 Zero Trust expected to apply to the parallel file system data path?**

07 The pricing matrix

The companion workbook SPECTRE1_Pricing_Matrix.xlsx is structured to §1's stated requirements: breakouts for all node types, all three media strategies, adapter and

media capacity options, and lead times against every line. Yellow cells are inputs; everything else is a formula.

TAB	PURPOSE	BUILT-IN CHECKS
README	Legend, the RFP's pricing asks, and the hard constraints to check every line against.	—
Config Summary	Roll-up across All-HDD / All-Flash / Hybrid: capacity, power, racks, endpoints, total evaluated price, \$ per usable PB.	Flags capacity outside 150–250 PB, total power over 138 kW, per-rack power over 17.3 kW, endpoints over 120.
Node & SSU Pricing	Line-item pricing per node type, tagged by config, with U, watts, lead time, TAA origin and FIPS status.	Auto-totals U, kW and extended price; Config Summary sums by config tag.
Media Options	Drive-by-drive options with recording technology, SED and FIPS status, country of origin, \$/TB and endurance.	Automatically flags any HAMR / MAMR / SMR line as NON-COMPLIANT , plus NO SED and TAA FAIL.
Network Adapters	400 GbE / Slingshot / InfiniBand / Ultra Ethernet options with per-port price, watts and lead time.	Extended price roll-up.
TDS Pricing	Per-TDS build across all eleven component classes, x 2 systems.	Per-TDS and total roll-up.
Support & Maintenance	Years 1–4 base plus 5th-year, 6th-year and deferred options across fourteen support line items.	Per-year and per-item totals.
Services & Deliverables	Twenty-four deliverables from the Site Prep Plan through the full ATO and SCRM artifact set, each with its RFP reference.	Total price and total hours.
Lead Time & Risk	Ten known long-lead components with drivers pre-filled; you add quoted lead times and mitigations.	—
Compliance Matrix	All 63 requirements pre-populated from Attachment C with section references. Mark Comply / Exception / Alternative on each.	\$3 requires every Exception to carry a discrepancy description, justification and mitigation — those columns are mandatory.

App. A The no's — and why

Every option we ruled out, with the reasoning recorded. If any of the assumptions below change — particularly a POSIX clarification or a budget increase — several of these decisions should be reopened.

A.1 Platforms we are not bidding

NO VAST Data

REOPEN IF Q4 ANSWERS FAVORABLY

Genuinely excellent at small files and QLC efficiency, and on the §4.1 distribution the technology would perform well. Two requirements land against it.

POSIX §3.2.2 requires "a single POSIX-compliant namespace." VAST has no native POSIX client — access is NFS, SMB and S3. NFSv4.1 with RDMA and nconnect gets close, but for MPI-IO across a Slingshot or InfiniBand fabric, POSIX semantics over NFS is a substantive technical difference, not a marketing one.

CONTAINERS §3.2.3 states a preference against container orchestrators in the data path and control plane. VAST's DASE architecture runs its CNodes as Docker containers. No Kubernetes is required, so it is arguable — but it means arguing against the grain of a stated preference while also arguing the POSIX point.

Decision: do not bid unless ORNL answers Q4 in favor of NFS-delivered POSIX semantics. Losing one of those two arguments is survivable; losing both is not.

NO Pure Storage FlashBlade //S and //E

CATEGORY MISMATCH

FlashBlade is an excellent unstructured-data platform, but it is not an HPC parallel file system. Namespace scale at 150–250 PB, POSIX semantics, and per-endpoint parallel bandwidth at $120 \times 400\text{G}$ are all outside what the architecture targets. //E is capacity-optimized for a different workload profile entirely — nearline and backup, not aged scratch under IOR and mdtest.

Decision: no. Bidding it would fail §3.2.2 and §4 on the technical evaluation regardless of price.

NO Dell PowerScale (OneFS)

CATEGORY MISMATCH

OneFS is a scale-out NAS with NFS and SMB access, not a parallel file system with a native POSIX client. It cannot deliver parallel client bandwidth at this endpoint count, and the same §3.2.2 POSIX argument that blocks VAST applies here with less compensating performance.

Decision: no.

NO Qumulo · NetApp · general object storage

CATEGORY MISMATCH

None of these are HPC scratch file systems. Object storage fails §3.2.2 outright — a bucket namespace is not a POSIX namespace, and no amount of gateway layering makes it one for MPI-IO under acceptance test conditions.

Decision: no.

NO BeeGFS (ThinkParQ)	SUPPORT DEPTH
Technically capable and used in parts of DoD HPC. The disqualifier is commercial, not technical: a six-year fixed-price PoP in a classified enclave, with a Q-cleared on-site analyst (§5.5), a two-week self-sufficient parts cache (§5.2), a 30-day CVSS v4 remediation SLA (§3.1.8), and the full 800-53 High artifact package (§3.1.13) demands a support organization of a scale BeeGFS does not have in the US federal market. Decision: no.	

NO Hammerspace — as the primary namespace	PARTIAL: OVERLAY ONLY
Parallel NFSv4.2 and the Tier 0 concept are interesting, and Hammerspace could add value as a data-orchestration overlay. But as the primary 150–250 PB scratch namespace carrying acceptance risk under §4.2, the deployment track record at this scale and in this environment does not support it. Decision: not as primary. Revisit as an overlay only if ORNL raises a specific data-movement requirement it solves.	

A.2 Architectural options we ruled out

NO All-HDD as the primary bid	STILL PRICED AS AN ALTERNATE
It reaches the capacity requirement — roughly 184 PB usable in the 8-rack envelope — and it is the cheapest path to raw petabytes. It fails on the other two axes simultaneously. POWER 87 JBODs at ~1,300 W consume 114 kW of a ~138 kW budget, leaving around 24 kW for the entire server layer. That funds roughly 20 × 400G endpoints against a permitted 120. SMALL FILES 72.3% of files are ≤64 KiB. Serving that population from spinning media, at 70% aged utilization, under mdtest, against published specifications, is the worst possible combination of §4.1 and §4.2. Decision: price it because §1 requires all three options, but do not lead with it. If a competitor leads with all-HDD on price, the power arithmetic in §03 is the argument that beats it.	

NO

All-flash as the primary bid

STILL PRICED AS AN ALTERNATE

This one is a genuinely close call, and the reason for the "no" is *not* technical. All-flash fits the facility envelope better than anything else: ~250 PB usable within the power budget, using about half the available rack space, supporting 48–64 endpoints.

COST 250 PB of QLC is on the order of 4,000 high-capacity drives. For a *scratch* file system this is very hard to justify against the alternative.

SCHEDULE §1 explicitly cites "unusual market conditions causing long-lead times." NAND allocation through 2026–27 is the specific risk the RFP is warning about, and delivery is Q1 CY2027.

ENDURANCE Scratch is write-heavy and churny. QLC DWPD against six years of scratch turnover needs modelling before anyone commits to it.

Decision: price it as the premium alternate and state the lead-time risk honestly. If ORNL answers Q1 such that 250 PB is required in 8 racks, this becomes the *only* compliant answer and we should be ready to pivot to it.

NO

Flash + object capacity tier (WEKA-style tiering)

ARCHITECTURAL

Tiering a flash namespace to an object capacity tier is how several vendors would close the capacity gap economically. It runs into §3.2.2's "single POSIX-compliant namespace" — a tiered architecture with an object backing store is defensible as one namespace but invites the argument — and it adds a second hardware footprint the 8-rack, 138 kW envelope has no room for.

Decision: no. If a vendor proposes it, the footprint and power must be counted in full, not treated as out-of-band.

NO

Maximizing endpoint count to the 120 ceiling

DESIGN CHOICE

§3.2.2 permits up to 120 HSN endpoints and it is tempting to bid to the ceiling as a differentiator. At ~1.1–1.2 kW per 400G server, 120 endpoints alone consume roughly 130 kW — essentially the entire 8-rack budget, before a single drive.

Decision: no. Right-size endpoints to the power budget, show the arithmetic, and surface the capacity-versus-bandwidth trade explicitly rather than letting the evaluator discover it. A bid that silently under-delivers endpoints looks worse than one that explains why 120 is not physically reachable.

A.3 Components ruled out

RULED OUT	REF	REASON
Seagate Mozaic 30 TB+ (HAMR)	§3.2	HAMR explicitly not permitted.

RULED OUT	REF	REASON
WD UltraSMR 32 TB / 40 TB-class	§3.2	Shingled recording explicitly not permitted.
Toshiba MAMR high-capacity line	§3.2	MAMR explicitly not permitted.
Any HDD without a FIPS SED variant	§3.2	FIPS 140-2 certified or 140-3 compliant SEDs required on all storage media.
Passive copper DAC between cabinets	§3.1.4	Inter-cabinet cabling must be optical or shielded twisted pair.
Mixed DIMM types across node classes	§3.2	ECC DIMMs must be a uniform FRU type system-wide.
Appliance platforms that lock the OS	§3.1.10, §5.4	DISA STIG compatibility and RHEL licensing both require the OS be modifiable without voiding support.
Raised-floor or under-floor distribution designs	§7	On-grade slab, no raised floor; all connectivity overhead.
TDS hardware used as the spare parts cache	§5.2	Explicitly prohibited. Spares must be priced separately.

Basis. Attachment C, *SPECTRE-1 Scratch Storage System Technical Requirements* v0.4 (May 2026); Attachment A (Terms & Conditions); Attachment B (Prime Flowdown Clauses); Attachment F draft subcontracts for Hardware and for OEM Maintenance & Support.

Estimates. Capacity, power and inode figures are engineering estimates derived from the stated assumptions and public vendor specifications. Re-derive them against vendor-confirmed part numbers before submission — in particular the 26 TB FIPS SED question in §3.1, which moves every capacity number in this brief.

Product data. WD Ultrastar DC HC590 and Data102 specifications; Solidigm D5-P5336 FIPS 140-3 capability; DDN AI400X3 / AI400X3M (ISC 2026); HPE Cray Storage Systems E2000; IBM Storage Scale System 6000; VDURA V5000 with WD Data60/Data102 shelves. All figures are vendor-published and should be confirmed in writing as part of the RFP response.

Competition sensitive — internal capture use only.